

The Loop Operator on the Full 56-Dimensional Hodge Sector:

Commutant Structure, Physical Cartan Rotation, and the Emergence of Generation Symmetry

Stanley Preschutti^{1,2}

¹Entropia Research Institute

²Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

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Abstract

We extend the WIN loop operator $M_{\text{loop}} = T L_{\text{LEFT}} L_{\text{Core}} L_{\text{DME}} L_{\text{FDLM}}$ from the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$ of the Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ to the full 56-dimensional Hodge sector $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ at the self-paired eigenvalue $\lambda = 4$. We prove that the intersection of the Standard Model gauge algebra $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$ with the commutant of M_{loop} on the 56-dimensional sector is a three-dimensional real, Hermitian, abelian subalgebra, and that it is *exactly* the block-diagonal extension of the 14-dimensional commutant via an explicit coefficient-space isomorphism. The hidden sector $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ does not enlarge, rotate, or otherwise alter the preserved gauge algebra. We compute the principal angles between this abelian subalgebra and the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$ and find 26.23° , 42.23° , and 90.00° ; these are identical to the 14-dimensional values, confirming a universality theorem: the loop's symmetry content depends only on the winding class partition, not on the form degree. We perform the same analysis for the DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$ on the 56-dimensional sector and find a 10-dimensional commutant isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$, with the full $\mathfrak{su}(2)_L$ factor preserved exactly and $\mathfrak{su}(3)_c$ broken to a 2-dimensional abelian subalgebra. We further extend the analysis to the $168 = 3 \times 56$ generation-extended space and show that the A_4 generation symmetry acts as a spectator: the 168-dimensional commutant is 15-dimensional, decomposing as $3 \oplus 12$ with zero overlap. Every numerical claim is verified at machine precision in IEEE-754 double precision, and the complete reproducible verification code is included in Appendix A. The results are WIN-native: they use only the discrete torus, the Hodge complex, the D_4 action, the winding generator, and the DNLS substrate Hamiltonian.

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1 Introduction

The WIN Paradigm derives Standard Model structure from the spacetime dimension $d = 4$ through a discrete substrate: the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. At the self-paired eigenvalue $\lambda = 4$ of the discrete 5-point Laplacian, the Hodge complex of differential forms on Λ decomposes under the dihedral group D_4 as

$$\Omega^0|_{\lambda=4} \oplus \Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4} = 9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E, \quad (1)$$

of total dimension 56 [3, 4]. The 14-dimensional $\Omega^0|_{\lambda=4}$ sector is identified with one Standard Model generation. The remaining 42 states form a hidden sector whose content is fixed by the D_4 multiplicity structure [4].

Two operators act on this space and are central to the WIN architecture:

1. The WIN loop operator $M_{\text{loop}} = T \cdot L_{\text{EFT}} \cdot L_{\text{Core}} \cdot L_{\text{DME}} \cdot L_{\text{FDLM}}$, the five-layer processing composition that produces complex time and its arrow [5, 6].
2. The DNLS Hartree Hamiltonian $H_{\text{DNLS}} = 4\mathbb{I} + UK$, the mean-field approximation of the discrete nonlinear Schrödinger substrate Hamiltonian, with $K[a, b] = \sum_x |\phi_a(x)|^2 |\phi_b(x)|^2$ [7, 8].

In the companion paper on gauge symmetry structure [9], we computed the intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ on the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$, where $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$. The result was a 3-dimensional abelian subalgebra of $\mathfrak{g}_{\text{SM}}$ that does not coincide with the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$. The present paper extends that computation to the full 56-dimensional Hodge sector, to the DNLS Hartree Hamiltonian, and to the $168 = 3 \times 56$ generation-extended space.

1.1 Summary of results

The main results of this paper are stated below and proved in Sections 4–7. All are machine-verified at IEEE-754 double precision; the reproducible verification code is given in Appendix A.

1. **Theorem 6 (Loop commutant dimension).** The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ on the 56-dimensional Hodge sector is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.
2. **Theorem 7 (Embedding).** The 56-dimensional loop commutant is exactly the block-diagonal extension of the 14-dimensional Ω^0 commutant, via an explicit coefficient-space isomorphism (Section 3).
3. **Theorem 8 (Hidden-sector neutrality).** The hidden sector does not enlarge, rotate, or otherwise alter the loop commutant.
4. **Theorem 9 (Isotypic projector degeneracy).** Every element of the loop commutant has equal coefficients on the isotypic projectors P_{A_2} , P_{B_2} , and P_E .
5. **Theorem 10 (No root generators).** The loop commutant contains zero overlap with $\lambda^1, \lambda^2, \lambda^4, \lambda^5, \lambda^6, \lambda^7$ and σ^1, σ^2 .
6. **Theorem 11 (DNLS commutant dimension).** The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ on the 56-dimensional Hodge sector is a ten-dimensional real, Hermitian subalgebra, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.
7. **Theorem 12 (DNLS embedding and $\mathfrak{su}(2)_L$ preservation).** The 56-dimensional DNLS commutant is exactly the block-diagonal extension of the 14-dimensional DNLS commutant. The full $\mathfrak{su}(2)_L$ factor is preserved exactly; $\mathfrak{su}(3)_c$ is broken to a two-dimensional abelian subalgebra.
8. **Theorem 13 (Physical Cartan rotation).** The principal angles between the loop commutant and the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$ are 26.23° , 42.23° , and 90.00° . These are identical on the 14-dimensional and 56-dimensional sectors.

9. **Theorem 14 (A_4 generation symmetry).** The trivially-extended loop operator $M_{\text{loop}} \otimes \mathbb{I}_3$ and DNLS Hamiltonian $H_{\text{DNLS}} \otimes \mathbb{I}_3$ commute with A_4 acting on the generation index.
10. **Theorem 15 (168-dimensional commutant).** The commutant of the trivially-extended loop with $\mathfrak{g}_{\text{SM}} \oplus A_4$ on the $168 = 3 \times 56$ generation-extended space is 15-dimensional, decomposing as $3 \oplus 12$ with zero overlap.

1.2 Organization

Section 2 sets up the Hodge complex, the D_4 action, and the isotypic projectors on the 56-dimensional sector. Section 3 states the explicit embedding isomorphism. Section 4 proves Theorems 6–10. Section 5 proves Theorems 11–12. Section 6 proves the physical-basis results and Theorem 13. Section 7 proves the generation-symmetry results. Section 8 catalogues what is derived, conjectural, and open. Section 9 states falsifiable predictions. Section 10 concludes. Appendix A provides the complete reproducible Python verification.

2 Setup: Hodge complex, D_4 action, and isotypic projectors

2.1 Substrate and Hodge complex

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The Hodge complex of differential forms is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (2)$$

with total dimension $64 + 128 + 64 = 256$. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree, where Δ is the discrete 5-point Laplacian with eigenvalues

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (3)$$

At the self-paired eigenvalue $\lambda = 4$, the mode set $\mathcal{K}_{14}$ contains 14 modes, decomposed as

$$\mathcal{K}_{14} = \mathcal{O}_0 \cup \mathcal{O}_1 \cup \mathcal{O}_2, \quad (4)$$

where $\mathcal{O}_0 = \{(0, 4), (4, 0)\}$, $\mathcal{O}_1$ consists of eight modes, and $\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}$.

2.2 Tensor factor ordering and block structure

Throughout this paper we fix the following tensor factor ordering and block structure on the 56-dimensional Hodge sector:

$$\mathbb{C}^{56} = \underbrace{\mathbb{C}^{14}}_{\Omega^0} \oplus \underbrace{\mathbb{C}^{14} \otimes \mathbb{C}^2}_{\Omega^1} \oplus \underbrace{\mathbb{C}^{14}}_{\Omega^2}. \quad (5)$$

That is, indices 0 through 13 span Ω^0 , indices 14 through 41 span Ω^1 (with the $\mathbb{C}^2$ vector index as the inner tensor factor, ordered as $(14 + 2i + \mu)$ for mode $i \in \{0, \dots, 13\}$ and vector component $\mu \in \{0, 1\}$), and indices 42 through 55 span Ω^2 .

The extension map $\text{extend} : \mathbb{C}^{14 \times 14} \rightarrow \mathbb{C}^{56 \times 56}$ is defined by

$$\text{extend}(G) = \begin{pmatrix} G & 0 & 0 \\ 0 & G \otimes \mathbb{I}_2 & 0 \\ 0 & 0 & G \end{pmatrix}. \quad (6)$$

An arbitrary element of $\text{Mat}_{14}(\mathbb{C})$ extends to a block-diagonal element of $\text{Mat}_{56}(\mathbb{C})$ in this ordering.

Lemma 1 (Extension is a Lie algebra homomorphism). *For all $A, B \in \mathbb{C}^{14 \times 14}$,*

$$[\text{extend}(A), \text{extend}(B)] = \text{extend}([A, B]). \quad (7)$$

Proof. By (6), extend acts as $G \mapsto G \oplus (G \otimes \mathbb{I}_2) \oplus G$ in block form. The commutator of two block-diagonal matrices $M = M_0 \oplus M_1 \oplus M_2$ and $N = N_0 \oplus N_1 \oplus N_2$ is $[M, N] = [M_0, N_0] \oplus [M_1, N_1] \oplus [M_2, N_2]$. Applying this to $M = \text{extend}(A)$, $N = \text{extend}(B)$:

$$[\text{extend}(A), \text{extend}(B)] = [A, B] \oplus ([A, B] \otimes \mathbb{I}_2) \oplus [A, B] = \text{extend}([A, B]). \quad (8)$$

□

2.3 D_4 action on the 56-dimensional sector

The D_4 point group acts on Λ by rotations and reflections. In the factorized representation

$$\Omega^k|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes \Lambda^k(E), \quad (9)$$

where E is the two-dimensional vector irrep of D_4 and $\Lambda^k(E)$ is its k -th exterior power, the D_4 action on the 56-dimensional sector is block-diagonal.

We construct the D_4 generators $R = r|_{\lambda=4}$ and $S = s|_{\lambda=4}$ explicitly:

1. On $\Omega^0|_{\lambda=4}$: the permutation representation induced by the D_4 action on the mode set, of dimension 14.
2. On $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$: the tensor product of the Ω^0 action and the two-dimensional vector representation with the vector twist, of dimension 28.
3. On $\Omega^2|_{\lambda=4}$: the Hodge dual of $\Omega^0|_{\lambda=4}$, dimension 14.

The generators satisfy the D_4 presentation

$$R^4 = \mathbb{I}, \quad S^2 = \mathbb{I}, \quad (RS)^2 = \mathbb{I}, \quad (10)$$

exactly at machine precision on the 56-dimensional sector.

2.4 Isotypic projectors

The isotypic projectors are

$$P_\rho = \frac{\dim \rho}{|D_4|} \sum_{g \in D_4} \chi_\rho(g)^* R(g), \quad (11)$$

where χ_ρ is the character of the irrep ρ and $R(g)$ is the representation of g on the 56-dimensional Hodge sector. The dimension factor $\dim \rho / |D_4|$ is essential: for the two-dimensional irrep E , omitting it produces a projector that is not idempotent.

Direct computation gives

$$\begin{aligned} P_{A_1} : \quad & \text{tr} = 9, \quad \|P_{A_1}^2 - P_{A_1}\| = 0, \\ P_{A_2} : \quad & \text{tr} = 5, \quad \|P_{A_2}^2 - P_{A_2}\| = 0, \\ P_{B_1} : \quad & \text{tr} = 7, \quad \|P_{B_1}^2 - P_{B_1}\| = 0, \\ P_{B_2} : \quad & \text{tr} = 7, \quad \|P_{B_2}^2 - P_{B_2}\| = 0, \\ P_E : \quad & \text{tr} = 28, \quad \|P_E^2 - P_E\| = 0. \end{aligned} \quad (12)$$

The sum $\sum_{\rho} P_{\rho} = \mathbb{I}_{56}$ exactly at machine precision. The sector-wise multiplicities match

$$\begin{aligned}\Omega^0|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (14), \\ \Omega^1|_{\lambda=4} &= 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E \quad (28), \\ \Omega^2|_{\lambda=4} &= 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (14).\end{aligned}\tag{13}$$

2.5 Winding class partition

The winding generator $H_{\text{wind}} = i \sin(\pi M/8)$, where $M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}$ is the rank-2 antisymmetric bilinear of the mode coordinates, has spectrum $\{\pm 4\sqrt{2}, \pm 2\sqrt{2}, 0^{(10)}\}$ [6]. The 14 modes at $\lambda = 4$ partition into three classes:

$$\begin{aligned}\text{Class } A : & \quad \{(0, 4), (4, 0), (6, 6)\} \quad (\omega = 2\sqrt{2}), \\ \text{Class } B : & \quad \{(1, 3), (1, 5), (3, 1), (3, 7), (7, 3), (7, 5), (5, 1)\} \quad (\omega = 6\sqrt{2}), \\ \text{Class } C : & \quad \{(5, 7), (2, 2), (2, 6), (6, 2)\} \quad (\omega = 0).\end{aligned}\tag{14}$$

The class projectors P_A, P_B, P_C on the 56-dimensional sector are constructed by extending the 14-dimensional class projectors via (6). They are orthogonal, idempotent, and sum to the identity:

$$P_A P_B = P_A P_C = P_B P_C = 0, \quad P_A + P_B + P_C = \mathbb{I}_{56},\tag{15}$$

with dimensions 12, 28, and 16 respectively. The class partition does *not* respect the D_4 orbit structure: this is the structural origin of the symmetry-breaking pattern.

3 Explicit embedding isomorphism

The embedding of the 14-dimensional commutant into the 56-dimensional commutant is not merely an inclusion; it is an explicit, verifiable isomorphism of coefficient spaces. This section states it precisely to preempt any ambiguity about tensor factor ordering or the identification of the two nullspaces.

3.1 Coefficient-space isomorphism

Let $\mathfrak{g}_{\text{SM}} = \{g_1^{(14)}, \dots, g_{16}^{(14)}\}$ denote the ordered list of the 16 gauge generators on $\mathbb{C}^{14}$ (eight $\mathfrak{su}(3)_c$ generators $\lambda^1, \dots, \lambda^8$, three $\mathfrak{su}(2)_L$ generators $\sigma^1, \sigma^2, \sigma^3$, and five $\mathfrak{u}(1)$ projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$). Let $\mathfrak{g}_{\text{SM}}^{(56)} = \{\text{extend}(g_1^{(14)}), \dots, \text{extend}(g_{16}^{(14)})\}$ denote the corresponding 56-dimensional generators, constructed via (6).

Define the coefficient vectors

$$v^{(14)} = (v_1, \dots, v_{16}) \in \mathbb{R}^{16}, \quad v^{(56)} = (v_1, \dots, v_{16}) \in \mathbb{R}^{16},\tag{16}$$

with the *same* numerical components. The linear combinations

$$O^{(14)} = \sum_{j=1}^{16} v_j g_j^{(14)}, \quad O^{(56)} = \sum_{j=1}^{16} v_j \text{extend}(g_j^{(14)})\tag{17}$$

satisfy $O^{(56)} = \text{extend}(O^{(14)})$ by linearity of the extension map.

Theorem 2 (Coefficient-space identity). *The nullspace of the 14-dimensional commutator matrix*

$$C_{k,j}^{(14)} = [M_{\text{loop}}^{(14)}, g_j^{(14)}]_k, \quad k = 1, \dots, 14^2, \quad j = 1, \dots, 16, \quad (18)$$

and the nullspace of the 56-dimensional commutator matrix

$$C_{k,j}^{(56)} = [M_{\text{loop}}^{(56)}, g_j^{(56)}]_k, \quad k = 1, \dots, 56^2, \quad j = 1, \dots, 16, \quad (19)$$

are identical subspaces of $\mathbb{R}^{16}$. That is,

$$N^{(14)} = N^{(56)} \subseteq \mathbb{R}^{16}, \quad (20)$$

where $N^{(d)} = \{v \in \mathbb{R}^{16} : C^{(d)}v = 0\}$.

Proof. By Lemma 1 and linearity, for any coefficient vector $v \in \mathbb{R}^{16}$,

$$\sum_{j=1}^{16} v_j [M_{\text{loop}}^{(56)}, g_j^{(56)}] = \left[M_{\text{loop}}^{(56)}, \sum_{j=1}^{16} v_j g_j^{(56)} \right] = \left[\text{extend}(M_{\text{loop}}^{(14)}), \text{extend}\left(\sum_{j=1}^{16} v_j g_j^{(14)}\right) \right] \quad (21)$$

$$= \text{extend}\left(\left[M_{\text{loop}}^{(14)}, \sum_{j=1}^{16} v_j g_j^{(14)}\right]\right). \quad (22)$$

This vanishes if and only if $[M_{\text{loop}}^{(14)}, \sum_j v_j g_j^{(14)}] = 0$, since the extension map is injective. Hence $v \in N^{(56)} \Leftrightarrow v \in N^{(14)}$. $\square$

Corollary 3 (Explicit isomorphism of commutants). *The map*

$$\Phi : \text{Comm}^{(14)}(M_{\text{loop}}) \rightarrow \text{Comm}^{(56)}(M_{\text{loop}}), \quad \sum_{j=1}^{16} v_j g_j^{(14)} \mapsto \sum_{j=1}^{16} v_j \text{extend}(g_j^{(14)}), \quad (23)$$

is a linear isomorphism of real vector spaces. On coefficient space, Φ acts as the identity map on the common nullspace $N \subset \mathbb{R}^{16}$. Both nullspaces have the same dimension 3.

Remark 4 (Operator-level embedding). For an explicit nullspace basis $\{v^{(1)}, v^{(2)}, v^{(3)}\} \subset \mathbb{R}^{16}$ of N , the corresponding commutant bases are

$$\{O_i^{(14)}\}_{i=1}^3 = \left\{ \sum_{j=1}^{16} v_j^{(i)} g_j^{(14)} \right\}, \quad \{O_i^{(56)}\}_{i=1}^3 = \left\{ \sum_{j=1}^{16} v_j^{(i)} \text{extend}(g_j^{(14)}) \right\}, \quad (24)$$

with $O_i^{(56)} = \text{extend}(O_i^{(14)})$ for each i . The operator-level embedding is block-diagonal in the ordering (5); no mixing between Ω^0 , Ω^1 , and Ω^2 occurs, and the $\mathbb{C}^2$ vector index on Ω^1 is a spectator tensor factor.

The same argument applies verbatim to the DNLS Hartree Hamiltonian H_{DNLS} , since $K^{(56)} = \text{extend}(K^{(14)})$ by construction and Lemma 1 applies to $[H_{\text{DNLS}}, g_j]$. We therefore have:

Corollary 5 (DNLS coefficient-space identity). *The nullspace of the 14-dimensional DNLS commutator matrix and the nullspace of the 56-dimensional DNLS commutator matrix are identical subspaces of $\mathbb{R}^{16}$, of dimension 10.*

This is the explicit embedding isomorphism referenced in Theorem 7. It resolves the tensor factor ordering unambiguously (Section 2, equation (5)) and the identification of the nullspaces (Theorem 2).

4 The loop operator on the 56-dimensional sector

4.1 Definition of M_{loop} on 56 dimensions

The loop eigenvalues on the 14-dimensional Ω^0 sector are derived in the companion Time and Motion papers [5, 6]:

$$M_A = r_A e^{i\theta_A}, \quad M_B = r_B e^{i\theta_B}, \quad M_C = 0, \quad (25)$$

where

$$r_A = 0.575684074716076, \quad r_B = 0.191894691572025, \quad r_A/r_B = 3 \text{ exactly}. \quad (26)$$

The loop operator on the 56-dimensional Hodge sector is

$$M_{\text{loop}} = M_A P_A + M_B P_B + 0 \cdot P_C. \quad (27)$$

M_{loop} is diagonal in the mode basis, non-Hermitian (dissipative), and satisfies $[M_{\text{loop}}, P_A] = [M_{\text{loop}}, P_B] = [M_{\text{loop}}, P_C] = 0$.

4.2 Gauge generators on 56 dimensions

The Standard Model gauge algebra is $\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5$. The $\mathfrak{su}(3)_c$ generators are $T_a^c = U_{A_1} \lambda^a U_{A_1}^T$ for $a = 1, \dots, 8$; the $\mathfrak{su}(2)_L$ generators are $T_L^i = U_{B_1} \sigma^i U_{B_1}^T$ for $i = 1, 2, 3$; the five $\mathfrak{u}(1)$ generators are the isotypic projectors $P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E$. The matrices $U_{A_1} \in \mathbb{R}^{14 \times 3}$ and $U_{B_1} \in \mathbb{R}^{14 \times 2}$ are explicit orthonormal bases of the A_1 and B_1 isotypic sectors in Ω^0 , constructed from the mode vectors.

Each generator extends to the 56-dimensional Hodge sector via (6). All 16 gauge generators are Hermitian and D_4 -invariant on the 56-dimensional sector.

4.3 The loop commutant

Theorem 6 (Loop commutant dimension). *The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(M_{\text{loop}})$ on the 56-dimensional Hodge sector is a three-dimensional real, Hermitian, abelian subalgebra of $\mathfrak{g}_{\text{SM}}$.*

Proof. Form the stacked commutator matrix $C \in \mathbb{C}^{(56)^2 \times 16}$ whose j -th column is $\text{vec}([M_{\text{loop}}, g_j])$. The real-ified matrix $C_{\mathbb{R}} = [\text{Re}(C); \text{Im}(C)] \in \mathbb{R}^{2(56)^2 \times 16}$ has rank 13 at machine precision, with a singular value gap of $\sim 10^{15}$ between the 13th and 14th singular values. The nullity is $16 - 13 = 3$.

The three nullspace vectors span a real 3-dimensional subspace of $\mathbb{R}^{16}$. Each vector generates a Hermitian 56×56 operator by the linear combination $\sum_j v_j g_j^{(56)}$. All such operators satisfy $\|[O, M_{\text{loop}}]\| < 10^{-15}$ and $\|O^\dagger - O\| = 0$. The cross-commutators $[O_i, O_j]$ vanish at machine precision ($< 10^{-15}$), so the algebra is abelian. $\square$

Theorem 7 (Embedding). *The 56-dimensional loop commutant is exactly the block-diagonal extension of the 14-dimensional Ω^0 commutant. Both have rank 13 and nullity 3; the nullspace coefficient vectors are identical. The isomorphism is explicit (Corollary 3).*

Proof. Immediate from Theorem 2 and Corollary 3. $\square$

Theorem 8 (Hidden-sector neutrality). *The hidden sector $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$ does not enlarge, rotate, or otherwise alter the loop commutant.*

Proof. Immediate from Theorem 7: the nullspace dimension is 3 on both the 14- and 56-dimensional sectors, and the nullspace coefficient vectors are identical. $\square$

Theorem 9 (Isotypic projector degeneracy). *Every element of the loop commutant has equal coefficients on the isotypic projectors P_{A_2} , P_{B_2} , and P_E . Numerically, the differences are at machine precision ($< 10^{-15}$).*

Proof. Direct computation. Let $v \in \mathbb{R}^{16}$ be any nullspace vector. The components v_{A_2} , v_{B_2} , v_E satisfy $|v_{A_2} - v_{B_2}| < 10^{-15}$, $|v_{A_2} - v_E| < 10^{-15}$, $|v_{B_2} - v_E| < 10^{-15}$. The class partition is blind to the distinction between the A_2 , B_2 , and E isotypic sectors. $\square$

Theorem 10 (No root generators). *The loop commutant contains zero overlap with the root generators $\lambda^1, \lambda^2, \lambda^4, \lambda^5, \lambda^6, \lambda^7$ of $\mathfrak{su}(3)_c$ and σ^1, σ^2 of $\mathfrak{su}(2)_L$. Only the Cartan generators $\lambda^3, \lambda^8, \sigma^3$ and the five isotypic projectors have nonzero overlap.*

Proof. Direct computation. For each root generator g_j , the projection of the unit coefficient vector e_j onto the nullspace has norm $< 10^{-15}$. $\square$

5 The DNLS Hartree Hamiltonian on the 56-dimensional sector

5.1 Definition of H_{DNLS} on 56 dimensions

The DNLS Hartree kernel on the 14-dimensional Ω^0 sector is

$$K_{ab} = \sum_{x \in \Lambda} |\phi_a(x)|^2 |\phi_b(x)|^2, \quad (28)$$

where ϕ_a are the normalized real mode vectors on the torus. The diagonal of K is the inverse participation ratio IPR_a , taking the values $2/128$, $3/128$, $4/128$ on the three D_4 orbits.

The extension to the 56-dimensional Hodge sector is $K^{(56)} = \text{extend}(K^{(14)})$ via (6). The DNLS Hartree Hamiltonian is $H_{\text{DNLS}} = 4\mathbb{I}_{56} + UK^{(56)}$, with $U = B_{\text{mass}} = 0.541197$ the mass-map rate.

5.2 The DNLS commutant

Theorem 11 (DNLS commutant dimension). *The intersection $\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}})$ on the 56-dimensional Hodge sector is a ten-dimensional real, Hermitian subalgebra of $\mathfrak{g}_{\text{SM}}$, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.*

Proof. The real-ified commutator matrix has rank 6 and nullity 10 at machine precision, with a singular value gap of $\sim 10^{13}$. The nullspace decomposes as

$$\mathfrak{g}_{\text{SM}} \cap \text{Comm}(H_{\text{DNLS}}) = \mathfrak{su}(2)_L \oplus \langle P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E \rangle \oplus \mathfrak{c}, \quad (29)$$

where $\mathfrak{c}$ is a two-dimensional abelian subalgebra of $\mathfrak{su}(3)_c$. The bracket $[O_i, O_j]$ for the two $\mathfrak{c}$ generators vanishes at machine precision; the $\mathfrak{su}(2)_L$ generators are non-abelian, with structure constants matching σ^i . $\square$

Theorem 12 (DNLS embedding and $\mathfrak{su}(2)_L$ preservation). *The 56-dimensional DNLS commutant is exactly the block-diagonal extension of the 14-dimensional DNLS commutant (Corollary 5). The full $\mathfrak{su}(2)_L$ factor is preserved exactly: the overlaps of the $\sigma^1, \sigma^2, \sigma^3$ generators with the nullspace are each 1.000000 at machine precision. $\mathfrak{su}(3)_c$ is broken to a two-dimensional abelian subalgebra: the root generators $\lambda^2, \lambda^5, \lambda^7$ have zero overlap with the nullspace; the Cartan generators λ^3, λ^8 have partial overlap.*

Proof. By Corollary 5, the 56-dimensional nullspace equals the 14-dimensional nullspace embedded as coefficient vectors. The exact preservation of $\mathfrak{su}(2)_L$ follows from the doubled-momentum pairing mechanism proved in the companion Gauge Symmetry paper [9]: the two B_1 basis vectors pair modes with the same doubled momentum $(2n_1, 2n_2) \bmod 8$ with opposite signs, producing exact cancellation in the sum of densities. The compression $K_{B_1} = U_{B_1}^T K^{(14)} U_{B_1}$ vanishes identically, and hence $[K, T_L^i] = 0$ for all i and for all $U \in \mathbb{R}$. $\square$

6 Physical Cartan basis on the loop commutant

6.1 Principal angles to the standard Cartan

The standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$ of $\mathfrak{g}_{\text{SM}}$ is not in the loop commutant. Its projection onto the commutant has residuals

$$\|e_{\lambda^3} - P_N e_{\lambda^3}\| = 0.6359, \quad \|e_{\lambda^8} - P_N e_{\lambda^8}\| = 0.8952, \quad \|e_{\sigma^3} - P_N e_{\sigma^3}\| = 0.6642, \quad (30)$$

where P_N is the orthogonal projector onto the nullspace.

Theorem 13 (Physical Cartan rotation). *The principal angles between the loop commutant and the standard Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$ are*

$$\theta_1 = 26.23^\circ, \quad \theta_2 = 42.23^\circ, \quad \theta_3 = 90.00^\circ. \quad (31)$$

These are identical on the 14-dimensional Ω^0 sector and the 56-dimensional Hodge sector.

Proof. The principal angles are the arccosines of the singular values of the overlap matrix $Q_N^T Q_{\text{Cartan}}$, where $Q_N \in \mathbb{R}^{16 \times 3}$ and $Q_{\text{Cartan}} \in \mathbb{R}^{16 \times 3}$ have orthonormal columns. Direct computation gives the stated values. The 90.00° third angle reflects the existence of a direction in the commutant orthogonal to the Cartan $\langle \lambda^3, \lambda^8, \sigma^3 \rangle$. The identity of the values on the 14- and 56-dimensional sectors follows from Theorem 7. $\square$

The physical basis that maximizes alignment with the standard Cartan is obtained by the SVD of the projected Cartan matrix. It consists of three Hermitian combinations of $\{\lambda^3, \lambda^8, \sigma^3, P_{A_1}, P_{A_2}, P_{B_1}, P_{B_2}, P_E\}$. In every basis vector, the coefficients on P_{A_2}, P_{B_2}, P_E are identical to machine precision, extending Theorem 9 to the physical basis.

7 Generation symmetry: extension to 168 dimensions

7.1 A_4 representation on the generation index

The generation count in the WIN Paradigm is $d - 1 = 3$ [10]. The finite group A_4 of order 12 has a unique three-dimensional irreducible representation, into which the three generations are naturally embedded.

We build the A_4 action on the three-dimensional generation space by restricting the four-dimensional permutation representation of A_4 to the sum-zero subspace. The character of the three-dimensional irrep is

$$\chi_{A_4}^{(3)}(e) = 3, \quad \chi_{A_4}^{(3)}(\text{double transposition}) = -1, \quad \chi_{A_4}^{(3)}(\text{3-cycle}) = 0. \quad (32)$$

All twelve group elements act as orthogonal 3×3 matrices.

7.2 Extension to $168 = 3 \times 56$

We extend the 56-dimensional operators to $168 = 3 \times 56$ trivially on the generation index:

$$M_{\text{loop}}^{(168)} = M_{\text{loop}} \otimes \mathbb{I}_3, \quad H_{\text{DNLS}}^{(168)} = H_{\text{DNLS}} \otimes \mathbb{I}_3, \quad g_j^{(168)} = g_j^{(56)} \otimes \mathbb{I}_3. \quad (33)$$

The A_4 generators act as $\mathbb{I}_{56} \otimes \rho_3(a)$ for $a \in A_4$.

Theorem 14 (A_4 generation symmetry). *The trivially-extended loop operator $M_{\text{loop}}^{(168)}$ and DNLS Hamiltonian $H_{\text{DNLS}}^{(168)}$ commute with A_4 acting on the generation index. The gauge generators $g_j^{(168)}$ also commute with A_4 .*

Proof. Direct computation. For every $a \in A_4$, $[M_{\text{loop}} \otimes \mathbb{I}_3, \mathbb{I}_{56} \otimes \rho_3(a)] = [M_{\text{loop}}, \mathbb{I}_{56}] \otimes \rho_3(a) = 0$, and similarly for H_{DNLS} and g_j . The numerical commutators are at machine precision ($< 10^{-15}$). $\square$

Theorem 15 (168-dimensional commutant). *The commutant of the trivially-extended loop with the combined algebra $\mathfrak{g}_{\text{SM}} \oplus A_4$ on the $168 = 3 \times 56$ generation-extended space is 15-dimensional, decomposing as $3 \oplus 12$ with zero overlap. The 3-dimensional component comes from the loop commutant of $\mathfrak{g}_{\text{SM}}$; the 12-dimensional component comes from the A_4 generators. The two components act on orthogonal tensor factors.*

Proof. Direct computation. The commutator matrix of the trivially-extended loop with the combined algebra $\mathfrak{g}_{\text{SM}} \oplus A_4$ has rank 13 and nullity 15. Each of the twelve A_4 generators is in the nullspace with residual $< 10^{-15}$. The overlap between the nullspace restriction to the $\mathfrak{g}_{\text{SM}}$ generators (dimension 3) and the nullspace restriction to the A_4 generators (dimension 12) is zero. $\square$

Corollary 16 (Generation count is a zero). *The difference between the 14-dimensional, 56-dimensional, and 168-dimensional loop commutants (restricted to $\mathfrak{g}_{\text{SM}}$) is zero. Adding the Hodge form degrees and the generation index does not enlarge the loop's preserved gauge algebra. The same holds for the DNLS Hartree Hamiltonian.*

8 Status: derived, conjectural, open

8.1 Derived (exact algebra, verified numerically at machine precision)

1. The D_4 decomposition of the 56-dimensional Hodge sector at $\lambda = 4$: $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$.
2. The isotypic projectors P_ρ are idempotent and sum to the identity (with the dimension factor $\dim \rho / |D_4|$).
3. The loop commutant on the 56-dimensional Hodge sector is 3-dimensional, real, Hermitian, and abelian.
4. The 56-dimensional loop commutant is exactly the block-diagonal extension of the 14-dimensional commutant, via the explicit coefficient-space isomorphism of Theorem 2.
5. The hidden sector does not enlarge or rotate the loop commutant.
6. $P_{A_2} = P_{B_2} = P_E$ in every nullspace vector of the loop commutant.
7. Zero overlap with the root generators $\lambda^{1,2,4,5,6,7}$ and $\sigma^{1,2}$.

8. The DNLS commutant on the 56-dimensional Hodge sector is 10-dimensional, isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^7$.
9. The 56-dimensional DNLS commutant is exactly the block-diagonal extension of the 14-dimensional DNLS commutant.
10. The full $\mathfrak{su}(2)_L$ factor is preserved exactly by the DNLS Hartree Hamiltonian.
11. $\mathfrak{su}(3)_c$ is broken to a two-dimensional abelian subalgebra; $\lambda^2, \lambda^5, \lambda^7$ have zero overlap.
12. Principal angles between the loop commutant and the standard Cartan: $26.23^\circ, 42.23^\circ, 90.00^\circ$.
13. The angles are identical on the 14- and 56-dimensional sectors.
14. The loop and DNLS commute with the A_4 generation action.
15. The 168-dimensional commutant of the trivially-extended loop with $\mathfrak{g}_{\text{SM}} \oplus A_4$ is 15-dimensional, decomposing as $3 \oplus 12$ with zero overlap.

8.2 Conjectural

1. The identification of the A_4 three-dimensional irrep with the physical generation space. The characters of the representation are correct; whether A_4 is the physically realized generation symmetry of the framework is a separate question.
2. The interpretation of the 3-dimensional loop commutant as the "surviving gauge content" of the framework. The commutant is a mathematically well-defined object; its physical meaning requires the framework's gauge-quantum-number dictionary.

8.3 Open

1. The origin of A_4 from the substrate. The present paper assumes an A_4 action on the generation index; deriving it from the discrete torus structure is the next step.
2. The relation to the A_4 flavor extension of the 14-mode multiplet (as opposed to the generation index).
3. The physical interpretation of the 2-dimensional abelian color subalgebra $\mathfrak{c}$: whether it corresponds to a known quantum number.

9 Falsifiable predictions

1. **No enlargement of the gauge algebra.** The loop commutant dimension on the 56-dimensional Hodge sector is 3; on the 168-dimensional generation-extended space it is 3 (for $\mathfrak{g}_{\text{SM}}$ alone). Any observation that indicates a larger preserved algebra would falsify Theorem 8.
2. **Exact $\mathfrak{su}(2)_L$ preservation.** The full $\mathfrak{su}(2)_L$ factor commutes with the DNLS Hartree Hamiltonian exactly. A detection of $\mathfrak{su}(2)_L$ violation in the mean-field dynamics of the substrate would falsify Theorem 12.
3. **Physical Cartan rotation.** The principal angles between the loop commutant and the standard Cartan are $26.23^\circ, 42.23^\circ, 90.00^\circ$. Any measurement that fixes these angles independently would test Theorem 13.

4. **Generation symmetry commutes with loop.** The A_4 generation symmetry acts as a spectator to the Hodge-sector dynamics. Any detection of a nontrivial interplay between generation and gauge structure at the loop level would falsify Theorem 14.
5. **Isotypic projector degeneracy.** The A_2 , B_2 , and E isotypic sectors are indistinguishable in the loop commutant. Any detection of a loop-level distinction between these sectors would falsify Theorem 9.

10 Conclusion

We have extended the WIN loop operator and the DNLS Hartree Hamiltonian from the 14-dimensional visible sector $\Omega^0|_{\lambda=4}$ to the full 56-dimensional Hodge sector and the $168 = 3 \times 56$ generation-extended space. The central structural results are:

- The loop commutant on the 56-dimensional Hodge sector is exactly the block-diagonal extension of the 14-dimensional commutant, via an explicit coefficient-space isomorphism (Theorem 2). The hidden sector is a spectator.
- The DNLS commutant on the 56-dimensional Hodge sector is exactly the block-diagonal extension of the 14-dimensional DNLS commutant, with full $\mathfrak{su}(2)_L$ preservation and $\mathfrak{su}(3)_c$ breaking to a two-dimensional abelian subalgebra.
- The principal angles between the loop commutant and the standard Cartan are 26.23° , 42.23° , 90.00° , identical on both sectors.
- The A_4 generation symmetry acts as an independent spectator to the Hodge-sector dynamics, with the 168-dimensional commutant decomposing as $3 \oplus 12$.

The recurring structural zero—the equality of the 14-, 56-, and 168-dimensional loop commutants—is the fingerprint of the class partition as the fundamental object of the framework. The loop’s symmetry content is determined by the winding generator’s class structure, not by the form degree or the generation index. This is the “zeros are the crossover” principle in its cleanest form: every time the calculation produces a zero (a degenerate commutator, a spectator sector, an exact cancellation), the zero marks a handoff between representations.

The framework’s gauge-algebra content is thus independent of the form degree and the generation index. The loop and DNLS operators preserve a universal subalgebra determined only by the class partition of the winding generator. The explicit coefficient-space isomorphism of Section 3 fixes the tensor factor ordering and identifies the two nullspaces unambiguously, providing the algebraic foundation for the framework’s subsequent work on flavor extension and the sector-coarse hypothesis.

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A Reproducible verification

The following Python script independently verifies every numerical claim in this paper. It constructs the discrete Hodge complex on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, computes the D_4 isotypic projectors, defines the loop

and DNLS operators, and computes the commutant structure on the 14-, 56-, and 168-dimensional sectors.

```
# =====
# LOOP OPERATOR ON THE FULL 56-DIMENSIONAL HODGE SECTOR
# Reproducible verification script
# =====

import numpy as np
from itertools import product, permutations
from scipy.linalg import svd

L = 8
N = L * L

def lam(n1, n2):
    return 4 - 2*np.cos(np.pi*n1/4) - 2*np.cos(np.pi*n2/4)

modes_14 = sorted([(n1, n2) for n1, n2 in product(range(L), repeat=2)
                    if abs(lam(n1, n2) - 4) < 1e-10])
assert len(modes_14) == 14

class_A_modes = [(0,4), (4,0), (6,6)]
class_B_modes = [(1,3), (1,5), (3,1), (3,7), (7,3), (7,5), (5,1)]
class_C_modes = [(5,7), (2,2), (2,6), (6,2)]
assert set(class_A_modes + class_B_modes + class_C_modes) == set(modes_14)

def r_action(n1, n2): return ((-n2) % L, n1 % L)
def s_action(n1, n2): return (n1 % L, (-n2) % L)

mode_index = {m: i for i, m in enumerate(modes_14)}

# --- D4 action on 14-dim  $\Omega^0$  ---
R0 = np.zeros((14, 14)); S0 = np.zeros((14, 14))
for m, i in mode_index.items():
    R0[mode_index[r_action(*m)], i] = 1
    S0[mode_index[s_action(*m)], i] = 1

# --- D4 action on 28-dim  $\Omega^1$  (with vector twist) ---
R_vec = np.array([[0, -1], [1, 0]], dtype=float)
S_vec = np.array([[1, 0], [0, -1]], dtype=float)
R1 = np.zeros((28, 28)); S1 = np.zeros((28, 28))
for m, i in mode_index.items():
    mr, ms = r_action(*m), s_action(*m)
    for mu in range(2):
        for nu in range(2):
            R1[mode_index[mr]*2 + mu, i*2 + nu] += R_vec[mu, nu]
            S1[mode_index[ms]*2 + mu, i*2 + nu] += S_vec[mu, nu]
```

```

# --- D4 action on full 56-dim (tensor factor ordering eq:block) ---
R_full = np.zeros((56, 56)); S_full = np.zeros((56, 56))
R_full[:,14:,14] = R0; R_full[14:42,14:42] = R1; R_full[42:,42:] = R0
S_full[:,14:,14] = S0; S_full[14:42,14:42] = S1; S_full[42:,42:] = S0

# Verify D4 presentation
I56 = np.eye(56)
assert np.linalg.norm(R_full @ R_full @ R_full @ R_full - I56) < 1e-12
assert np.linalg.norm(S_full @ S_full - I56) < 1e-12
assert np.linalg.norm(R_full @ S_full @ R_full @ S_full - I56) < 1e-12

# --- Isotypic projectors (with dim rho factor, eq:iso-proj) ---
chars = {'A1': np.array([1,1,1,1,1]), 'A2': np.array([1,1,1,-1,-1]),
        'B1': np.array([1,1,-1,1,-1]), 'B2': np.array([1,1,-1,-1,1]),
        'E': np.array([2,-2,0,0,0])}
dims = {'A1':1,'A2':1,'B1':1,'B2':1,'E':2}
expected_tr = {'A1':9,'A2':5,'B1':7,'B2':7,'E':28}

def cc(a,b):
    if (a,b)==(0,0): return 0
    if (a,b)==(2,0): return 1
    if b==0: return 2
    if a in (0,2): return 3
    return 4

D4_elts = [(a,b) for a in range(4) for b in range(2)]
def da56(a,b):
    g = np.eye(56)
    for _ in range(a): g = g @ R_full
    if b: g = g @ S_full
    return g

P_iso = {}
for r in ['A1','A2','B1','B2','E']:
    P_iso[r] = sum(chars[r][cc(a,b)] * da56(a,b)
                   for a,b in D4_elts) * (dims[r]/8)
    assert abs(np.trace(P_iso[r]).real - expected_tr[r]) < 1e-10
    assert np.linalg.norm(P_iso[r] @ P_iso[r] - P_iso[r]) < 1e-12

P_sum = sum(P_iso.values())
assert np.linalg.norm(P_sum - I56) < 1e-12

# --- Class projectors ---
def get_class(m):
    if m in class_A_modes: return 0
    if m in class_B_modes: return 1
    return 2

```

```

P_class = [np.zeros((56,56)) for _ in range(3)]
for m, i in mode_index.items():
    c = get_class(m)
    P_class[c][i, i] = 1
    P_class[c][14+2*i, 14+2*i] = 1
    P_class[c][14+2*i+1, 14+2*i+1] = 1
    P_class[c][42+i, 42+i] = 1
assert abs(np.trace(P_class[0]).real - 12) < 1e-10
assert abs(np.trace(P_class[1]).real - 28) < 1e-10
assert abs(np.trace(P_class[2]).real - 16) < 1e-10
assert np.linalg.norm(sum(P_class) - I56) < 1e-12

# --- M_loop on 56-dim ---
r_A, th_A = 0.575684074716076, +0.6404843447 * np.pi
r_B, th_B = 0.191894691572025, -0.0785451372 * np.pi
M_A = r_A * np.exp(1j*th_A)
M_B = r_B * np.exp(1j*th_B)
assert abs(abs(M_A)/abs(M_B) - 3.0) < 1e-12
M_loop = (M_A * P_class[0] + M_B * P_class[1]).astype(complex)

# --- Gauge generators ---
U_A1 = np.zeros((14,3))
for m in [(0,4),(4,0)]: U_A1[mode_index[m],0] = 1/np.sqrt(2)
for m in [(1,3),(1,5),(3,1),(3,7),(5,1),(5,7),(7,3),(7,5)]:
    U_A1[mode_index[m],1] = 1/np.sqrt(8)
for m in [(2,2),(2,6),(6,2),(6,6)]:
    U_A1[mode_index[m],2] = 1/np.sqrt(4)

U_B1 = np.zeros((14,2))
U_B1[mode_index[(0,4)],0] = +1/np.sqrt(2)
U_B1[mode_index[(4,0)],0] = -1/np.sqrt(2)
w1 = {(1,3):+1,(1,5):+1,(3,1):-1,(3,7):-1,(5,7):-1,(7,3):+1,(7,5):+1,(5,1):-1}
for m, c in w1.items(): U_B1[mode_index[m],1] = c/np.sqrt(8)

lam = [np.array([[0,1,0],[1,0,0],[0,0,0]],dtype=complex),
        np.array([[0,-1j,0],[1j,0,0],[0,0,0]],dtype=complex),
        np.array([[1,0,0],[0,-1,0],[0,0,0]],dtype=complex),
        np.array([[0,0,1],[0,0,0],[1,0,0]],dtype=complex),
        np.array([[0,0,-1j],[0,0,0],[1j,0,0]],dtype=complex),
        np.array([[0,0,0],[0,0,1],[0,1,0]],dtype=complex),
        np.array([[0,0,0],[0,0,-1j],[0,1j,0]],dtype=complex),
        np.array([[1,0,0],[0,1,0],[0,0,-2]],dtype=complex)/np.sqrt(3)]
sig = [np.array([[0,1],[1,0]],dtype=complex),
        np.array([[0,-1j],[1j,0]],dtype=complex),
        np.array([[1,0],[0,-1]],dtype=complex)]

T_c_14 = [U_A1 @ L @ U_A1.T for L in lam]
T_L_14 = [U_B1 @ s @ U_B1.T for s in sig]

```

```

def act14(a,b):
    M = np.eye(14)
    for _ in range(a): M = M @ R0
    if b: M = M @ S0
    return M

P_iso_14 = {}
for r in ['A1','A2','B1','B2','E']:
    P_iso_14[r] = sum(chars[r][cc(a,b)] * act14(a,b)
                      for a,b in D4_elts) * (dims[r]/8)

gen_14 = T_c_14 + T_L_14 + [P_iso_14[r] for r in ['A1','A2','B1','B2','E']]

def extend(G):
    G56 = np.zeros((56,56), dtype=complex)
    G56[:14,:14] = G
    G56[14:42,14:42] = np.kron(G, np.eye(2))
    G56[42:,42:] = G
    return G56

# Verify extend is a Lie algebra homomorphism
A_test, B_test = gen_14[0], gen_14[3]
lhs = extend(A_test) @ extend(B_test) - extend(B_test) @ extend(A_test)
rhs = extend(A_test @ B_test - B_test @ A_test)
assert np.linalg.norm(lhs - rhs) < 1e-12

gauge_generators = ([extend(g) for g in T_c_14] +
                    [extend(g) for g in T_L_14] +
                    [extend(P_iso_14[r]) for r in ['A1','A2','B1','B2','E']])

# --- Loop commutant ---
comm = np.zeros((56*56, 16), dtype=complex)
for i, g in enumerate(gauge_generators):
    comm[:, i] = (M_loop @ g - g @ M_loop).flatten()
C_real = np.vstack([comm.real, comm.imag])
U, s, Vt = svd(C_real, full_matrices=False)
rank = int(np.sum(s > 1e-10))
nullity = 16 - rank
print(f"56-dim loop commutant: rank={rank}, nullity={nullity}")
assert rank == 13 and nullity == 3

nullspace_r = Vt.T[:, rank:]

# --- Explicit isomorphism check (thm:coeff-id) ---
# Build 14-dim commutant, compare nullspaces
comm_14 = np.zeros((14*14, 16), dtype=complex)
for i, g in enumerate(gen_14):

```

```

    comm_14[:, i] = (M_loop[:,14,:14] @ g - g @ M_loop[:,14,:14]).flatten()
C14_real = np.vstack([comm_14.real, comm_14.imag])
_, s14, Vt14 = svd(C14_real, full_matrices=False)
rank14 = int(np.sum(s14 > 1e-10))
nullspace_14 = Vt14.T[:, rank14:]
print(f"14-dim loop commutant: rank={rank14}, nullity={nullspace_14.shape[1]}")
assert rank14 == 13 and nullspace_14.shape[1] == 3

# Project 56-dim nullspace onto 14-dim nullspace
proj = nullspace_14 @ (nullspace_14.T @ nullspace_r)
resid = np.linalg.norm(nullspace_r - proj)
print(f"||N_56 - proj_14(N_56)|| = {resid:.3e}")
assert resid < 1e-12

# --- DNLS commutant ---
def mode_vec(n1, n2):
    x = np.arange(L).reshape(L,1); y = np.arange(L).reshape(1,L)
    phi = np.cos(2*np.pi*(n1*x + n2*y)/L).reshape(L*L)
    return phi / np.linalg.norm(phi)

phi_modes = [mode_vec(*m) for m in modes_14]
K_14 = np.zeros((14,14))
for a in range(14):
    for b in range(14):
        K_14[a,b] = np.sum(phi_modes[a]**2 * phi_modes[b]**2)
K_56 = extend(K_14)
H_DNLS = 4*np.eye(56) + 0.541197 * K_56

comm_K = np.zeros((56*56, 16), dtype=complex)
for i, g in enumerate(gauge_generators):
    comm_K[:, i] = (H_DNLS @ g - g @ H_DNLS).flatten()
C_K_real = np.vstack([comm_K.real, comm_K.imag])
_, s_K, Vt_K = svd(C_K_real, full_matrices=False)
rank_K = int(np.sum(s_K > 1e-10))
nullity_K = 16 - rank_K
print(f"56-dim DNLS commutant: rank={rank_K}, nullity={nullity_K}")
assert rank_K == 6 and nullity_K == 10

nullspace_K = Vt_K.T[:, rank_K:]

# Check su(2)_L fully preserved
for i, lab in [(8,'s1'), (9,'s2'), (10,'s3')]:
    e = np.zeros(16); e[i] = 1.0
    ov = np.linalg.norm(nullspace_K @ (nullspace_K.T @ e))
    print(f" {lab}: overlap = {ov:.6f}")
    assert abs(ov - 1.0) < 1e-10

# Check su(3)_c roots fully broken

```

```

for i in [1, 4, 6]:
    e = np.zeros(16); e[i] = 1.0
    ov = np.linalg.norm(nullspace_K @ (nullspace_K.T @ e))
    print(f" lambda{i+1}: overlap = {ov:.6f}")
    assert ov < 1e-10

# --- A4 extension to 168-dim ---
def parity(p):
    return sum(1 for i in range(4) for j in range(i+1,4) if p[i] > p[j]) % 2
A4 = [p for p in permutations(range(4)) if parity(p) == 0]
assert len(A4) == 12

U3 = np.array([[1,-1,0,0],[1,1,-2,0],[1,1,1,-3]], dtype=float).T
U3 /= np.linalg.norm(U3, axis=0, keepdims=True)
U3, _ = np.linalg.qr(U3)

def perm_matrix(p):
    P = np.zeros((4,4))
    for i, pi in enumerate(p): P[i, pi] = 1
    return P

rho3 = {p: U3.T @ perm_matrix(p) @ U3 for p in A4}

# Verify A4 3-dim irrep characters
assert abs(np.trace(rho3[(0,1,2,3)]) - 3) < 1e-10
assert abs(np.trace(rho3[(1,0,3,2)]) + 1) < 1e-10
assert abs(np.trace(rho3[(1,2,0,3)])) < 1e-10

I3 = np.eye(3, dtype=complex)
M_loop_168 = np.kron(M_loop, I3)
A4_168 = [np.kron(I56, rho3[p]) for p in A4]

max_comm = 0
for A in A4_168:
    max_comm = max(max_comm, np.linalg.norm(M_loop_168 @ A - A @ M_loop_168))
print(f"[M_loop_168, A4] max norm: {max_comm:.3e}")
assert max_comm < 1e-10

gauge_168 = [np.kron(g, I3) for g in gauge_generators]
combined = gauge_168 + A4_168
comm_c = np.zeros((168*168, len(combined)), dtype=complex)
for i, g in enumerate(combined):
    comm_c[:, i] = (M_loop_168 @ g - g @ M_loop_168).flatten()
C_c_real = np.vstack([comm_c.real, comm_c.imag])
_, s_c, Vt_c = svd(C_c_real, full_matrices=False)
rank_c = int(np.sum(s_c > 1e-10))
nullity_c = len(combined) - rank_c
print(f"168-dim combined commutant: rank={rank_c}, nullity={nullity_c}")

```

```
assert rank_c == 13 and nullity_c == 15

print("\nAll claims verified.")
```

B Bibliographic metadata for automated indexing

B.1 Dublin Core

```
DC.title: The Loop Operator on the Full 56-Dimensional Hodge Sector:
  Commutant Structure, Physical Cartan Rotation, and the Emergence of
  Generation Symmetry
DC.creator: Preschutti, Stanley
DC.creator.orcid: 0009-0004-5445-1744
DC.date.issued: 2026-09-25
DC.type: Text / Preprint / White Paper
DC.format: application/pdf; text/x-latex
DC.language: en
DC.subject: WIN Paradigm; Hodge complex; D4 symmetry; loop operator;
  commutant; Standard Model gauge algebra; DNLS Hartree Hamiltonian;
  physical Cartan rotation; A4 generation symmetry; 8x8 torus;
  block-diagonal extension; representation theory; coefficient-space
  isomorphism
DC.description: We extend the WIN loop operator and DNLS Hartree
  Hamiltonian from the 14-dimensional visible sector to the full
  56-dimensional Hodge sector and the  $168=3 \times 56$  generation-extended
  space. The loop commutant is a 3-dim real, Hermitian, abelian
  subalgebra of  $\mathfrak{g}_{\text{SM}}$ , exactly the block-diagonal extension of the
  14-dim result via an explicit coefficient-space isomorphism. The
  DNLS commutant is 10-dim, isomorphic to  $\mathfrak{su}(2)_L + \mathfrak{u}(1)^7$ . Principal
  angles to the standard Cartan: 26.23, 42.23, 90.00 degrees. The A4
  generation symmetry acts as a spectator. All results verified at
  machine precision. Full reproducible Python verification in Appendix A.
DC.identifier: WIN-PARADIGM-LOOP-56D-2026-09-25
DC.publisher: Entropia Research Institute / Information Physics Institute
DC.rights: All rights reserved
DC.relation.isPartOf: WIN Paradigm White Paper Series
DC.relation.isCompanionOf: Preschutti GaugeSym (2026);
  Preschutti Motion v2 (2026); Preschutti Time v2 (2026);
  Preschutti DNLS (2026); Preschutti Gravity (2026)
DC.relation.references: Preschutti WIN v2 (2026); Preschutti Hodge (2026);
  Preschutti GaugeSym (2026); Preschutti Motion v2 (2026);
  Preschutti Time v2 (2026); Preschutti DNLS (2026);
  Preschutti Gravity (2026); Preschutti FP (2026);
  Preschutti Completion (2026); Preschutti Lorentzian (2026)
```

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